RETROFIT AI

An operator assistant for the pumps you already have

Early, explainable, listen-only fault detection for existing cooling-water pumps:
no new machinery, no new controls, no write path.

Since AI Hackathon 2026, Turku | Challenge: Business Turku | Technical sparring: Elomatic

Old pumps fail quietly; alarms come late

Ships and plants run 10-25-year-old pumps with a handful of signals and fixed alarm limits

11 %

of developing problems on the validation pumps are caught by the plant's own alarm list (organizer baseline B0)

0.44

fault-type recall of the best simple baseline: half the time it names the wrong cause (B1, validation)

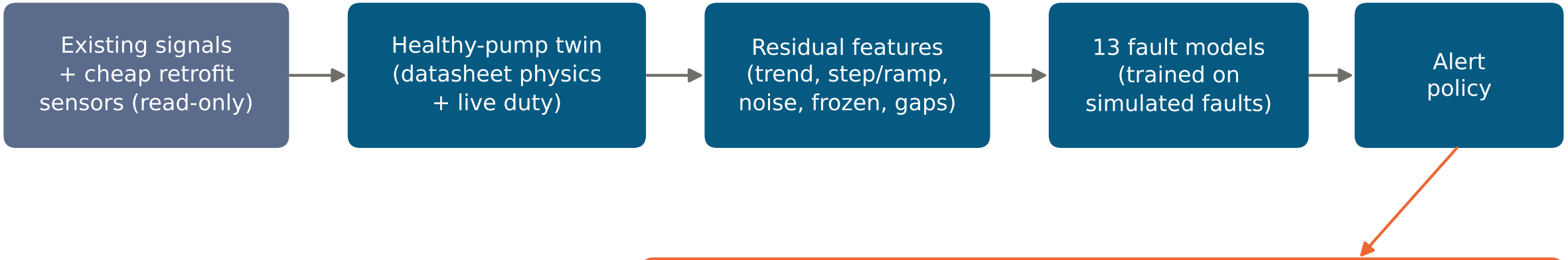
EUR 9 k

typical unplanned repair vs EUR 4 k planned, plus 10 h vs 3 h stop (organizer business-case assumptions)

Replacing working pumps or their controls to add AI is costly, so the intelligence has to be retrofitted around them.

Listen, compare with physics, diagnose, advise

Every minute, on an edge box next to the pump; nothing is ever written back



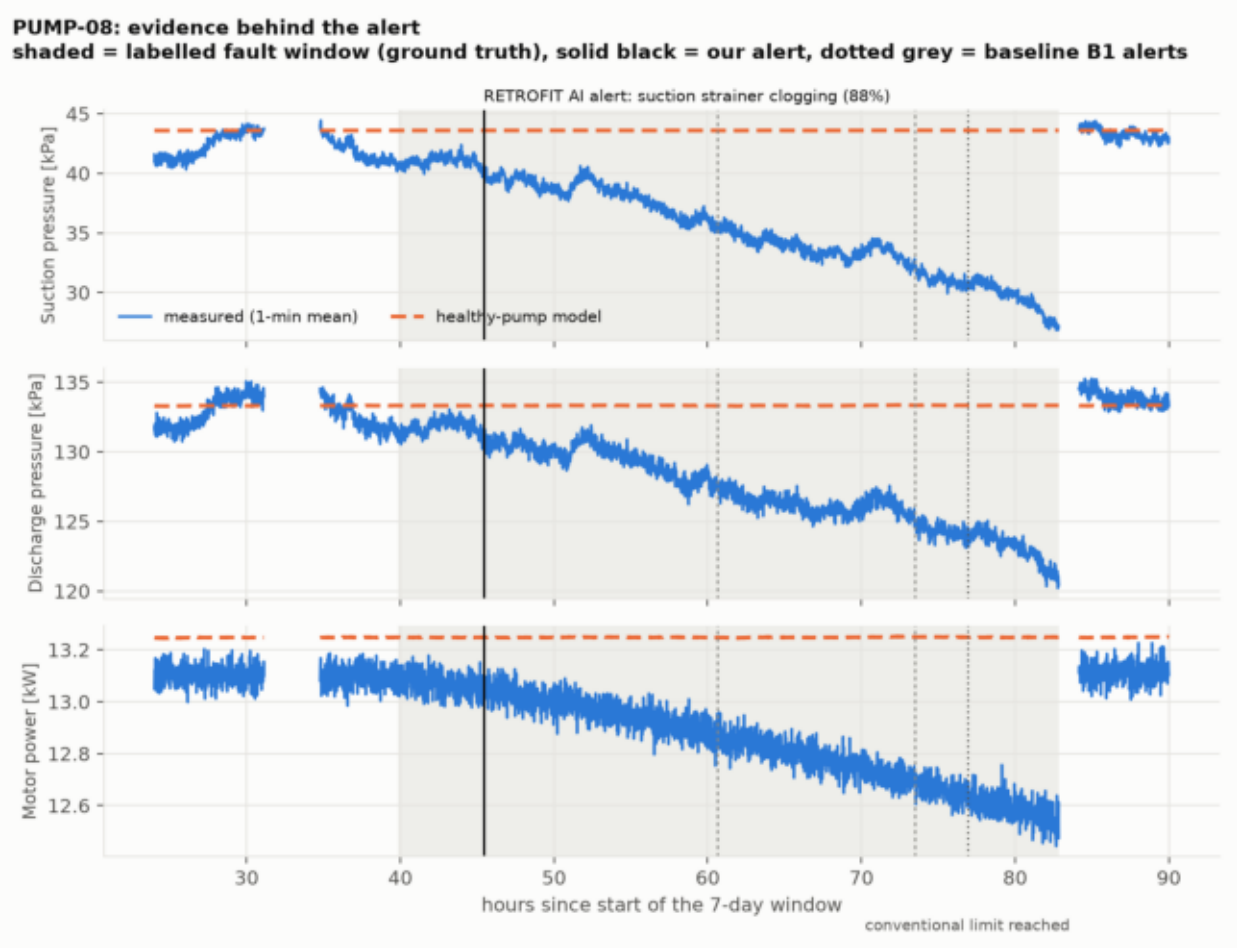
What is new

- reference = physics, not 'the first day of data':
  faults present at start are visible
- diagnosis learned from 500 simulated pump-weeks
  (organizers' simulator, all 13 fault types)
- one alert per problem, never a command

Advisory alert: machine or instrument fault, likely cause, confidence, evidence in kPa / % / mm/s, other condition indicated, recommended check -> a human approves (console, work-order request)

Evidence an operator can check

PUMP-08 (validation, wireless retrofit): suction strainer clogging



Alert card

Suction strainer clogging
(machine fault)

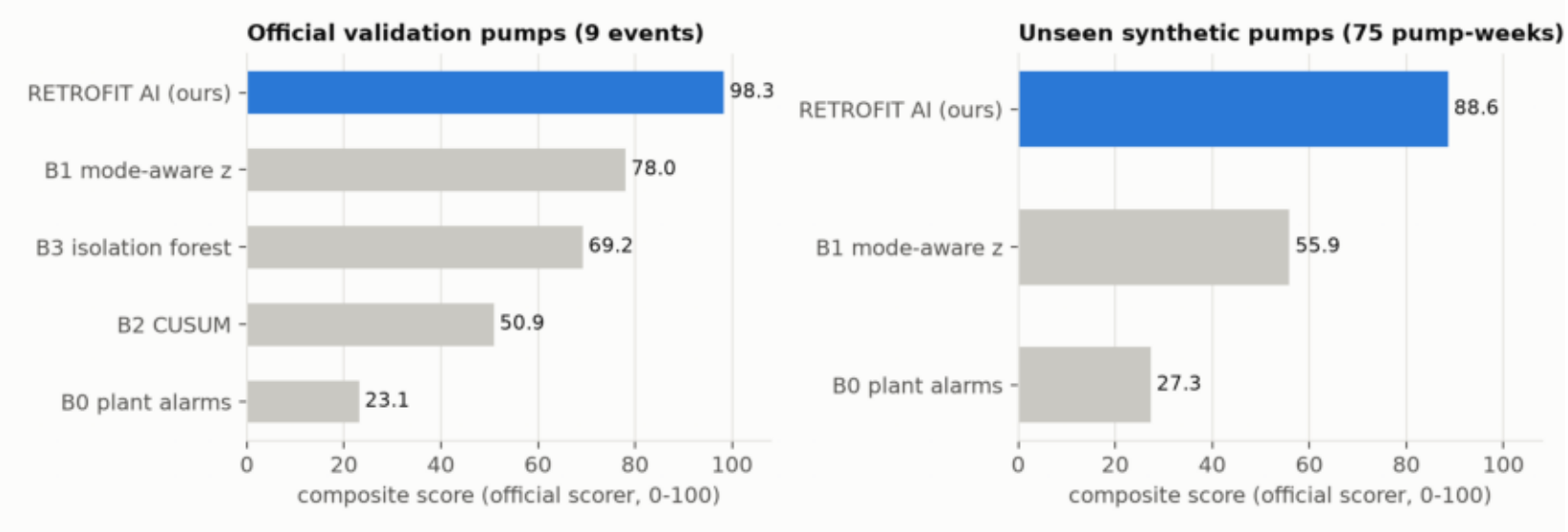
Evidence: suction pressure falls below the healthy model; discharge pressure and power fall with it (less flow)

Recommended (human approves): clean / backflush the strainer at the next opportunity

Severity estimate from physics:
clog factor ~2.2 -> sandbox: cleaning restores flow and NPSH margin

Measured with the official scorer

Composite = 35 % F1 + 20 % timeliness + 25 % fault-type + 10 % class + 10 % alert burden



100%

of validation faults named with the right cause (best baseline 44%)

84%

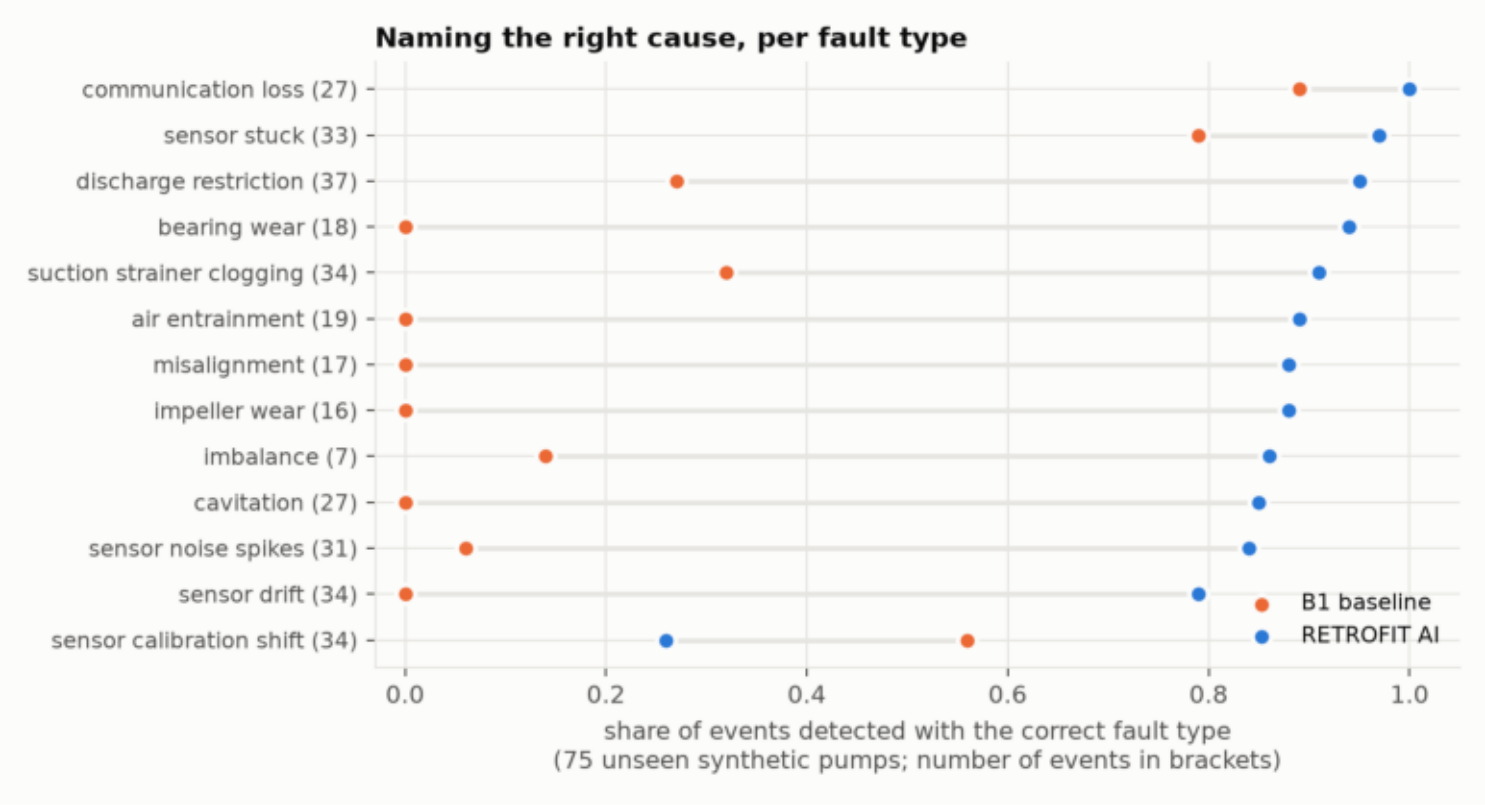
on 75 unseen simulated pumps (best baseline 28%)

Validation: false + excess alerts 0.0 per pump-day, timeliness 0.91. Hold-out composite 95 % bootstrap interval 86.4-90.7 (resampling pumps).

Validation pumps never used for training or tuning; thresholds tuned on separate simulated hold-out pumps.

It names the right cause, including 'it is the sensor'

Instrument faults change one reading while the physically coupled readings stay healthy



Instrument vs machine

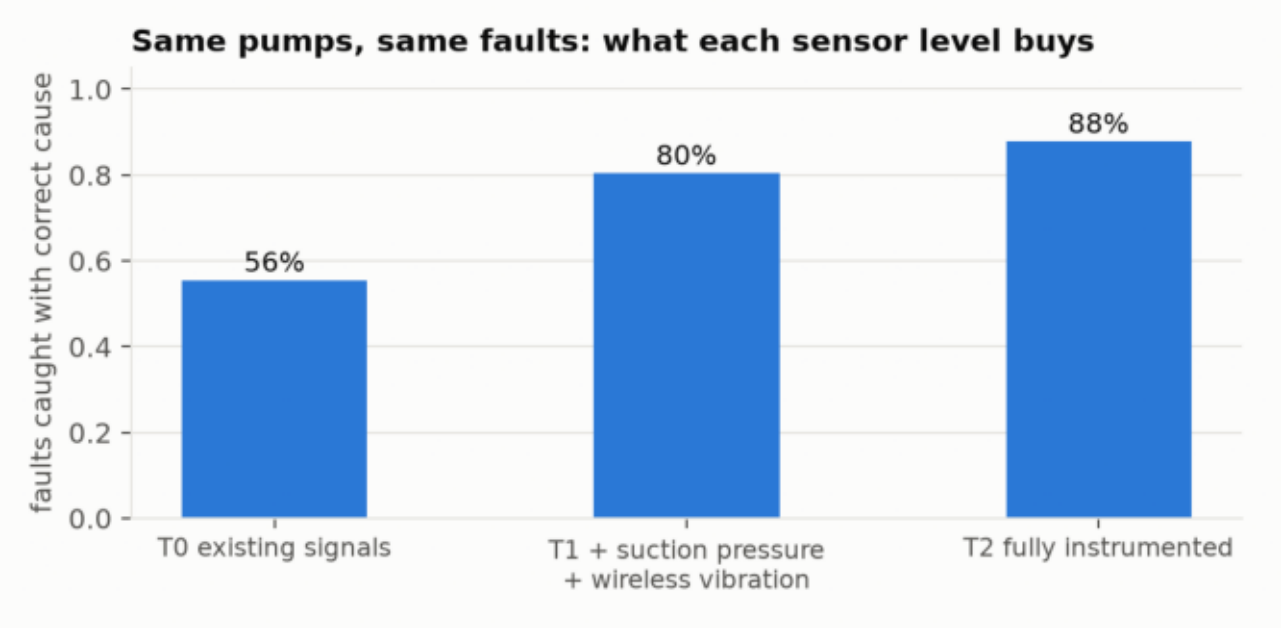
A drifting or frozen gauge can trigger a needless repair or hide a real fault.

The assistant checks each reading against the others through the pump physics (curve, affinity laws, motor):
- one channel off, peers healthy
  -> instrument fault, check transmitter
- several channels move as the physics predicts -> machine fault

Step vs ramp shape separates a calibration shift from a drift.

What each sensor buys, on identical pumps

Same simulated pumps and faults at three instrumentation levels



Retrofit bill of materials (list prices)

T0 existing signals: edge box only,
EUR 500-600 (RevPi / IOT2050)
T1 + suction transmitter (~EUR 145) + one wireless vibration point (EUR 300-1,500)
-> about EUR 1-2.5 k per pump
T2 full (IO-Link vibration, power, flow, PT100): EUR 2.1-3.1 k + flow meter

Recommendation: start every pump at T0; add the T1 pair on critical pumps without redundancy; T2 where energy metering pays.

Safe by design

Advisory only; existing alarms, trips and interlocks stay authoritative

No write path

Reads OPC UA / Modbus / replay; one-way MQTT-style alerts. IEC 62443 zones, IACS E26/E27.

Human approval

Acknowledge, reject, request work order; every decision logged for shadow review.

Sandbox only

What-if checks run on the simulator with duty, NPSH, load and vibration limits.

Honest uncertainty

Confidence, other condition indicated, low confidence shown as 'possible', blind spots stated.

Causal, audited

Truncated re-runs give identical alerts: PASSED (10 cut points).

Edge-sized

Median 259.3 ms per run, 341.9 MB peak, offline, no GPU, no cloud.

From hackathon to pilot

Value comes from avoided unplanned stops minus nuisance handling, measured in shadow mode

EUR 8,600

net per pump and year in the organizer's worked example, with our measured recall and false-alert rate (illustrative, synthetic)

0.5 years

simple payback on EUR 4.5 k installed (organizer assumptions)

1 Read-only data, 4-6 weeks: calibrate the healthy model

2 Shadow mode: log alerts, compare with work orders

3 Add one T1 point where blind spots show

4 Advisory pilot with a named role and accept/reject

5 Scale on measured value; independent review first

Transfers; with honest limits

Swap the physics layer, keep the rest

Other rotating machines

Fans and blowers: same affinity laws, filter clogging = strainer analogue
Compressors: map + surge margin instead of curve + NPSH margin
Motors, gearboxes: load model, gear-mesh frequency instead of vane pass

Reused unchanged: features, policy, console, edge runtime, sensor checks

Limits

Synthetic data, simulator-to-simulator: scores overstate field performance
Datasheet defaults need a commissioning baseline on a real pump
No vibration sensor -> no bearing alerts
9 validation events: wide uncertainty
Advisory, uncertified, no proven savings